

Understanding Pointwise vs. Uniform Convergence of Functions

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Normed Vector Spaces

Let E be a vector space. A *norm* on E is a function: $v \rightarrow |v|$ from E into $\mathbb{R}$ that satisfies the following axioms:

- 1 $|v| \geq 0$, and $|v| = 0 \iff v = 0$.
- 2 $\forall a \in \mathbb{R}, v \in E, |av| = |a||v|$.
- 3 $\forall v, w \in E, |v + w| \leq |v| + |w|$.

A vector space with a norm is called a normed vector space [Lang, 1983].

Supremum

For a set S , we say the set is bounded from above if there is a number c such that: $\forall x \in S, x \leq c$. We refer to c as a upper bound of S . Let R be the set of all upper bounds of S . We then define the least upper bound b as the minimum element of R . The element b is frequently called $\sup(S)$ [Lang, 1983].

Sup Norm

Let f be a bounded function on S . We define the sup norm of f ,
 $\|f\| = \sup_{x \in S} |f(x)|$.

Checking if this is a norm...

- 1 $\|f\| = 0 \implies |f(x)| = 0$ for all x . Also,
 $|f(x)| = 0 \implies \sup_{x \in S} |f(x)| = 0$.
- 2 consider f and g . $\|fg\| = \sup_{x \in S} |f(x)g(x)| =$
 $\sup_{x \in S} |f(x)| \sup_{x \in S} |g(x)| = \|f\| \|g\|$.
- 3 Let $\|f\| = M$, $\|g\| = N$. Then $\|f + g\| = \sup_{x \in S} |f + g| \leq$
 $\sup_{x \in S} |f| + \sup_{x \in S} |g| = \|f\| + \|g\|$.

Thus, the sup norm is a norm.

Using ϵ and N to prove convergence

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2 Let $n \geq N$.

3 "Discover" that this $n \geq N$ leads to the desired inequality
($\lim_{n \rightarrow \infty} |f_n - f_N| \leq \epsilon$).

Pointwise Convergence - Definition

Definition

Let S be a set, and F a normed vector space. Let $\{f_n\}$ be a sequence of maps: $S \rightarrow F$. For each $x \in S$ we may then consider the sequence of elements of F given by $\{f_1(x), f_2(x), \dots\}$. Thus for each x , we may speak of the convergence of the sequence $\{f_n(x)\}$. If $\{f_n\}$ is a sequence of maps such that for each $x \in S$ the sequence $\{f_n(x)\}$ converges, then we say that $\{f_n\}$ converges pointwise [Lang, 1983].

Pointwise Convergence - Example 1

$\lim_{x \rightarrow \infty} \frac{1}{n}x^2 = 0$ converges pointwise.

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Proof.

$\forall \epsilon > 0, \epsilon \in \mathbb{R}$ there exists an $N = \frac{x^2}{\epsilon}$ s.t. $\forall n > N$ we know $|\frac{1}{n}x^2| < |\frac{1}{\frac{x^2}{\epsilon}}x^2|$ as $n > \frac{x^2}{\epsilon}$. Thus $|\frac{1}{n}| < \frac{1}{\frac{x^2}{\epsilon}}x^2 = \frac{\epsilon}{x^2}x^2 = \epsilon$ for all x . □

Our N value is dependent on the x coordinate of the function.
Different $f(x)$ converge to 0 at different rates.

Pointwise Convergence - Example 2

Consider the sequence of functions:

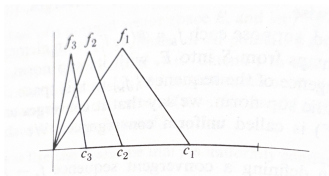


Figure: credit:[Lang, 1983]

Let $f_n(x) = 0$ when $x \geq c_n$, where c_n approaches zero. Then the limit of the functions will equal zero. So $\{f_n\}$ would converge pointwise to zero [Lang, 1983].

Sup Norm's role in Uniform Convergence

$$\begin{aligned} \lim_{n \rightarrow \infty} \{f_n\} &= f \text{ Uniformly} \\ \implies \lim_{n \rightarrow \infty} \sup\{|f_n(x) - f(x)| : \forall x\} &= 0 \\ \implies \lim_{n \rightarrow \infty} \|f_n(x) - f(x)\| &= 0 \end{aligned}$$

credit: Wikipedia, "Uniform convergence"

The Question to Ask

Recall the Algorithm:

- 1 Determine an equation for ϵ that depends on N (and maybe $x\dots$).
- 2 Let $n \geq N$.
- 3 "Discover" that this $n \geq N$ leads to the desired inequality $(\lim_{n \rightarrow \infty} |f_n - f_N| \leq \epsilon)$.

Now... does the ϵ equation depend on x ?

Uniform Convergence - Definition

Definition

Let S be a set, and F a normed vector space. Let $\{f_n\}$ be a sequence of maps: $S \rightarrow F$. For each $x \in S$ we may then consider the sequence of elements of F given by $\{f_1(x), f_2(x), \dots\}$.

Suppose each f_n is an element of the vector space of bounded maps: $f_n \in \beta(S, F)$ from S into F , with its sup norm. Then we may speak of the convergence of the sequence $\{f_n\}$ in this space. If the sequence $\{f_n\}$ converges for the sup norm, we say that it converges uniformly. Convergence in $\beta(S, F)$ is called uniform convergence. The sup norm is indicated using $\| \cdot \|$ [Lang, 1983].

Uniform Convergence - Example 1

$\lim_{n \rightarrow \infty} \frac{1}{n} \sin x = 0$ converges uniformly.

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Proof.

For a given $\epsilon > 0, \epsilon \in \mathbb{R}$ observe that $\forall n > \frac{1}{\epsilon}$ we know $||\frac{1}{n} \sin x|| \leq ||\frac{1}{n}|| = \frac{1}{n}$. Thus $||\frac{1}{n} \sin x|| \leq \frac{1}{n} < \frac{1}{\frac{1}{\epsilon}} = \epsilon$. □

Observe that our $N = \frac{1}{\epsilon}$ does not vary with respect to x , thus there is a uniformity to the rate of convergence.

Uniform Convergence - Example 2

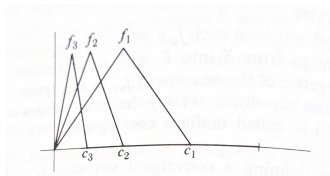


Figure: credit: [Lang, 1983]

Observe that, for all $n \in \mathbb{N}$, $\|f_n(x) - 0\| = 1$.

In this case, however, the supremum norm is a constant. This means that the sequence cannot converge uniformly to 0 as the supremum norm cannot be within an arbitrary ϵ of 0. [Lang, 1983].

Pointwise Implies Uniform

It is not difficult to show that any function which converges uniformly will also converge pointwise uniformly.

Uniform Limit Theorem

There are two major theorems in regards to uniform convergence. First we have the uniform limit theorem [Auliac and Caby, 2005].

Theorem

Let $\{f_n\}$ be a sequence of mappings from a metric space M to a metric space N . If $\forall n \in \mathbb{N} : f_n$ is continuous at every point of M and $\{f_n\}$ uniformly converges to f , then f is continuous at every point of M .

Uniform Limit Theorem - Example

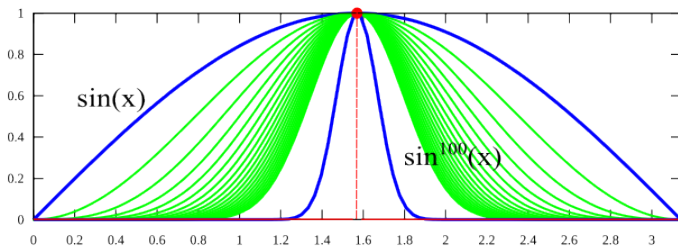


Figure: Continuous Function Sequence [Ika, 2013]

Riemann Integral

The second theorem has to do with the Riemann integral.

Theorem

If $\{f_n\}$ is a sequence of functions which uniformly converge to f then $\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b f$ where $\int$ is the Riemann integral.

Riemann Integration - Example

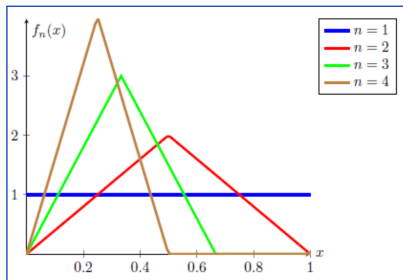


Figure: Integration of a Function Sequence [Murthy and Gavosto, 2023]

Questions?



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